

NIST Requirements Document
Title: Teledyne Camera Repair
Date: April 29, 2025

I. Background Information

Single-photon-level spectrum measurement and characterization are essential for the development and validation of quantum photonic devices used in quantum communication and networking. These techniques enable us to accurately assess key performance metrics such as emission wavelength, linewidth and coherence properties parameters of quantum devices like single-photon sources and entangled-photon sources. Currently, we are developing single/entangled source-based color-center based SiC device in collaboration with Morgan State University and Carnegie Mellon University (CMU). Once developed, these sources will be integrated into the quantum network testbeds at the National Institute of Standards and Technology (NIST), including both the NIST Gaithersburg Quantum Network Testbed (NG-QNet) and the Washington, DC-area testbed (DC-QNet). These testbeds are designed to evaluate real-world performance of quantum technologies and serve as a foundation for the next-generation quantum internet infrastructure.

NIST has a single-photon-level spectrum measurement system developed by Teledyne, which plays a critical role in our research on quantum photonic devices. This system is specifically designed to measure and characterize the photon spectra of quantum emitters, including single-photon and entangled-photon sources. Accurate spectral characterization at the single-photon level is essential for the research on such devices. A key component of the system is the PyLoN-IR InGaAs cooled camera, which is responsible for detecting low-intensity infrared photons with high sensitivity and low noise. As the camera recently stopped functioning and had to be returned to the manufacturer for evaluation, and after a thorough inspection by Teledyne, it was confirmed that the camera hardware had failed and requires repair. Given the importance of this camera to the ongoing quantum device characterization efforts, the NIST lab requires the necessary repairs of the camera so that critical measurements and continue progress on the quantum research projects can resume.

II. Purpose

The purpose of this requirement is to repair an existing PyLoN-IR InGaAs cooled camera (Model Number / Serial Number: Model#: HRS-500-SS, Model ID HRS500-SS-19-SS-NP-MS-NO-SF-SF-YI, S/N: 5000366) for Teledyne spectrum measurement system.

III. Specific Requirements

The contractor shall repair the PyLoN-IR InGaAs cooled camera for use with the Teledyne spectrum measurement system and ensure full functionality within that system.

IV. Delivery

Delivery shall be FOB DESTINATION and shall occur within eight (8) weeks or less after receipt of order (ARO). FOB Destination means: The contractor shall pack and mark the

shipment in conformance with carrier requirements, deliver the shipment in good order and condition to the point of delivery specified in the purchase order, be responsible for any loss of and/or damage to the goods occurring before receipt and acceptance of the shipment by the consignee at the delivery point specified in the purchase order; and pay all charges to the specified point of delivery. The contractor shall deliver the repaired camera to:

The National Institute of Standards and Technology
ATTN: Oliver Slattery, 100 Bureau Drive, Building 222 / A215, Gaithersburg, MD,
20899-1640

V. Inspection and Acceptance:

In addition to the inspection and acceptance terms articulated in 52.212-4, the Government reserves the right to perform such performance tests and evaluations as defined below to verify specified system performance. Such tests and evaluations, if performed, shall be conducted within the environment that the system is to be operated. The Contractor has the right to be present during the tests and evaluations, if performed, at the Contractor's expense.

The scope of general inspection included:

- Structural, mechanical, and electrical evaluation of product
- Evaluation of the camera's performance within the Teledyne spectrum measurement system

A visual inspection of the system will be performed by the NIST TPOC to identify defects or any form of indication that the system was damaged during transport to NIST. The Government shall have sole discretion to require repair or replacement of damaged and/or nonconforming supplies at no cost to the Government. The Government at any time prior to acceptance shall reject the system due to defects and/or nonconformance.

The vendor is responsible for latent defects discovered any time after final inspection. However, the extent of its liability shall be prorated over the useful life of the system.

The place of acceptance shall be:

100 Bureau Drive
Building 222/A215
Gaithersburg, MD 20899

VI. Warranty

The contractor shall extend the warranty for one (1) year after acceptance of the repair in accordance with the contractor's commercial warranty terms and shall be in accordance with terms in FAR 52.212-4.

VII. Payment Schedule

The Contractor shall be paid, in accordance with Net 30-day payment terms, upon receipt and acceptance of a proper invoice, in accordance with the following schedule:

1. 100% after receipt and acceptance by the TPOC of fully delivered system.
2. The Government anticipates inspection will occur:
 - a. After the testing procedures set forth in this document have been complete